



Bivariate Data Analysis Topic Test 1

Solution

Total Marks: 30

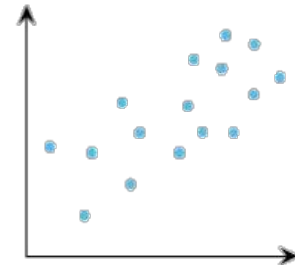
Time: 50 minutes

- 1 For the scatterplot is shown, which of the following best describes the correlation? 1

- A. Strong positive
- B. Weak positive
- C. Strong negative
- D. Weak negative

Weak Positive

B



- 2 Jemma is a political advisor who studies the effects of time on television over a month on the approval rating of six politicians. The data is shown below.

<i>Time (in minutes)</i>	20	10	70	15	5	80
<i>Approval rating (%)</i>	25	50	55	30	80	85

- a. Using a calculator, find Pearson's correlation coefficient rr . Answer correct to four 1

decimal places. Using the calculator $r = 0.40991611... \approx 0.4099$

- b. Jemma concludes that a new politician should appear on television as often as possible 1

to gain a high approval rating. Do you agree? Give a reason.

Disagree.

Pearson's correlation of 0.4099 indicates a weak positive correlation between time and television. In addition, Jack's investigation uses a small data sample.

- 3 The price of various second-hand Mazda cars are shown below.

Age (a)	1	2	0	5	7	4	3	4	8
Value (v)(£)	15000	11000	18900	9800	7500	11600	10500	11000	3500

- a. Find the equation of the least-square regression line in terms of Age (a) and Value (v) of the car. 1

$$A = 16747.2 \quad B = -1527.2$$

$$\therefore v = 16747.2 - 1527.2a$$

- b. What is the value of the y -intercept for this equation, and what does it represent in the context of the data? 2

y -intercept is 16747.20 (or 18900 from table)
The value of a new Mazda

- c. How many years will it take for the car to become worthless? 2

$$16747.2 - 1527.2a = 0$$

$$a = \frac{16747.20}{1527.20} = 10.9659... = 11 \text{ years}$$

- 4 A company is investigating its weight loss diet. They find 10 people who use their diet and recorded the time they've been on the diet and their weight loss.

Weeks on the diet	8	4	4	2	7	8	5	3	9	6
Weight loss (kg)	5	4	3	1	4	4	3	3	6	5

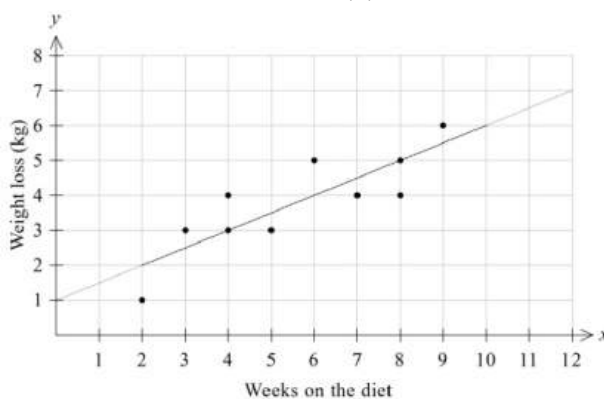
- a. Pearson's coefficient for this data is $r = 0.846$. What does this value say about the strength and direction of the correlation between the number of weeks on the diet and the weight loss? 1
- There is a strong positive relationship between the number of weeks on the diet and weight loss

- b. The least-squares regression line is $y=1+mx$, where: 1
- x is the number of weeks on the diet
- y is the weight loss, in kilograms
- m is the gradient of the line
- $m = 0.5$

Using technology, find the value of m .

- c. Interpret the value of m found in part (b) in the context of the data provided. 1
- On average, people lose 0.5 kg each week on the diet

- d. The scatter plot shows the data collected from the 10 people on the diet. Draw the least-square s regression line found in part (b). 2

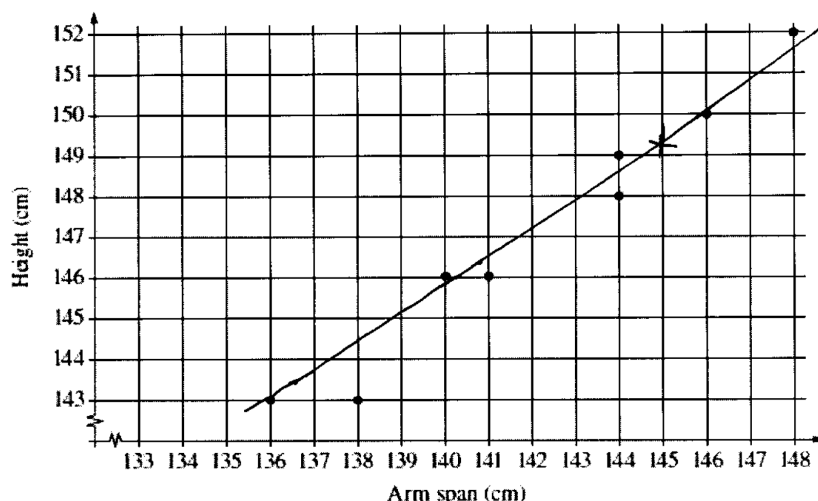


- e. Why does the y -intercept have no meaning in the context provided? 1
- The y -intercept of 1 would indicate people lose 1 kg after no weeks on the diet, which doesn't make sense.

- 5 A class compared their assessment results to their head circumference. The correlation coefficient for these quantities was 0.2. What is the meaning of this correlation? 1

Assessment results increase as head circumference increases.
Low positive correlation.
Not a strong relationship.

- 6 A set of bivariate data is collected by measuring the height and arm span of eight children. The graph shows a scatterplot of these measurements.



- a. On the graph, draw a line of best fit by eye. 1

- b. Tim is a child from the class who was absent when the measurements were taken. He has an arm span of 145 cm. Using your line of best fit from part (a), estimate Tim's height. 1

149.5

- c. Calculate the value of Pearson's correlation coefficient, correct to four decimal places. 2

Armspan: 136 | 138 | 140 | 141 | 144 | 144 | 146 | 148 |
 Height: 143 | 143 | 146 | 146 | 148 | 149 | 150 | 152 |
 $r = 0.9847$

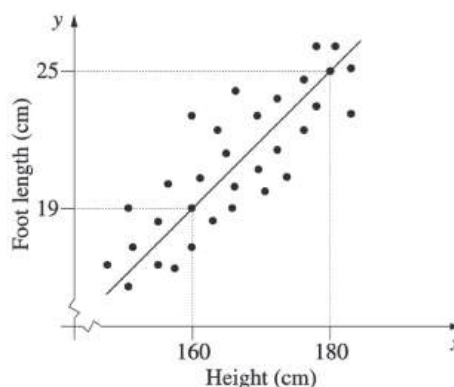
- d. Describe the strength of association between height and arm span. 1

positive association

- 7 Each member of a group of males had his height and foot length measured and recorded. The results were graphed and a line of fit drawn.

- a. Why does the value of the y-intercept have no meaning in this situation? 1

A person cannot have zero height



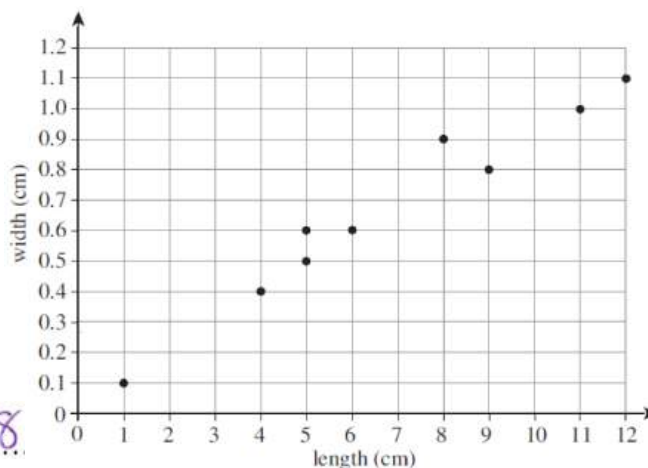
- b. George is 10 cm taller than his brother Harry. Use the line of fit to estimate the difference in their foot lengths. 1

difference is 3cm

- c. Sam calculated a correlation coefficient of -1.2 for the data. Give TWO reasons why Sam must be incorrect. 2

values of $-1 < r < 1$ and the correlation is positive not negative

- 8 A set of bivariate data is collected by measuring the lengths and widths of a population of slugs. A scatterplot of this data is shown.



- a. Calculate Pearson's correlation coefficient for the data, correct to two decimal places.

$$r = 0.98$$

1

- b. Identify the direction and strength of the linear relationship between the lengths and widths of the slugs.

strong, positive, linear relationship

2

- c. Find the equation of the least-squares regressions line in the form:

3

$$\text{width} = B \times \text{length} + A$$

and use the equation to predict the length of a slug that has a width of 0.3 cm correct to one decimal place.

$$A = 0.072$$

$$B = 0.088$$

$$\text{width} = 0.088 \text{ length} + 0.072$$

$$0.3 = 0.088 \times \text{length} + 0.072$$

$$\text{length} = 2.59 \text{ cm}$$



Bivariate Data Analysis Topic Test 2 Solution

Total Marks: 30

Time: 50 minutes

- 1 What is the best description for the relationship between a person's age and their life expectancy? 1

(D)

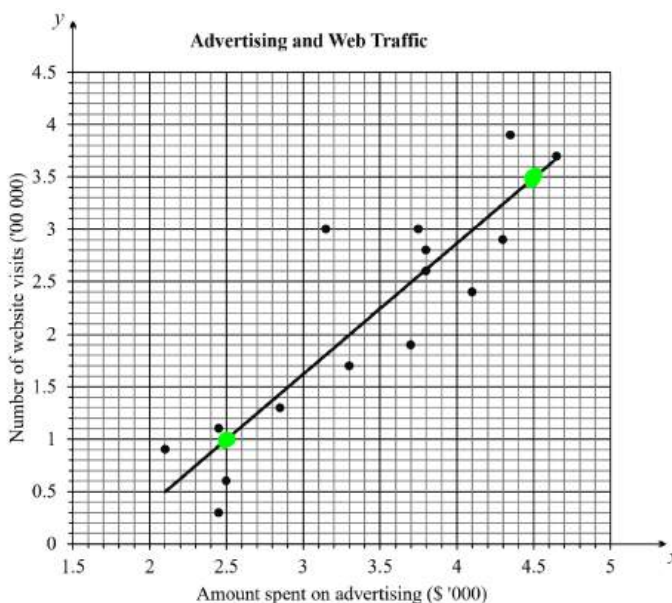
- A. Strong positive correlation
B. Weak positive correlation
C. Weak negative correlation
D. Strong negative correlation

- 2 A company wanted to investigate the strength of the relationship between the amount of money spent on advertising each week and the number of visitors the company's website receives each week. Data was collected over a 15 week period.

- a. Pearson's correlation coefficient for the data set is $r = 0.9$. Describe the strength and direction of this linear correlation. 1

strong positive

- b. The graph below shows a scatterplot and the regression line for the data. Use the graph to calculate the equation of the regression line.



$$\begin{aligned} m &= \frac{3.5 - 1}{4.5 - 2.5} = 1.25 \\ \therefore y &= 1.25x + b \\ (2.5, 1) \Rightarrow 1 &= 1.25(2.5) + b \\ b &= -2.125 \\ \therefore y &= 1.25x - 2.125 \end{aligned}$$

- c. Use the equation of the regression line from part (b) to predict the number of visitors the website will receive for a week in which \$3000 was spent on advertising. 1

$$\begin{aligned} y &= 1.25x - 2.125 \\ x &= 3, \quad y = 1.25(3) - 2.125 = 1.625 \\ \therefore \text{Will receive } 162500 \text{ visitors.} \end{aligned}$$

- d. Explain why the model is not useful for predicting the number of visitors for weeks in which \$1500 was spent on advertising. 1

As the model will give a negative amount of visitors.

- 3 School analysis of data suggests that there is a correlation between how a student performs in year 10 and their final HSC scores. Below are the results of 11 students in Mathematics in Year 10 and HSC.

Year 10 Mathematics Marks	69	87	78	54	80	89	53	50	69	68	90
HSC Mathematics Marks	39	46	42	37	43	46	34	35	39	40	47

- a. Calculate the value of the correlation coefficient. $r = 0.9806159..$ 1

- b. Describe the strength and direction of the correlation between these marks. 1

Strong, positive correlation.

- c. Find the equation of the least-squares regression line for the data. 2

$$y = 0.3x + 19.26$$

- d. Use your equation from part (c) to estimate an HSC mark for a student who achieves 75 in Year 10 Mathematics. 1

$$y = 0.3 \times 75 + 19.26$$

$$y = 41.76$$

- e. Is it certain that this student will achieve the HSC mark found in part (d) based on their 1

Year 10 result? Give a reason for your answer.

It is not certain. Other factors can influence a student's academic achievement between Year 10 and HSC such as application.

- 4 The Mathematics and Chemistry exam results for a group of students are recorded in the scatterplot. A line of best fit was drawn as shown.

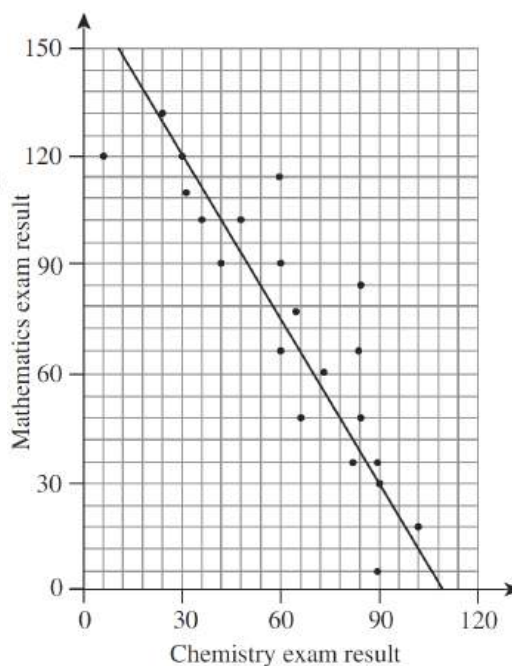
- a. Calculate the gradient of the line of best fit. 1

Using two points on the line ((30, 120) and (90, 30)) gives:

$$m = \frac{y_2 - y_1}{x_2 - x_1} = \frac{30 - 120}{90 - 30} = -\frac{3}{2} \text{ OR } -1.5$$

- b. Identify the correlation between the Mathematics and Chemistry exam results. 2

strong negative correlation



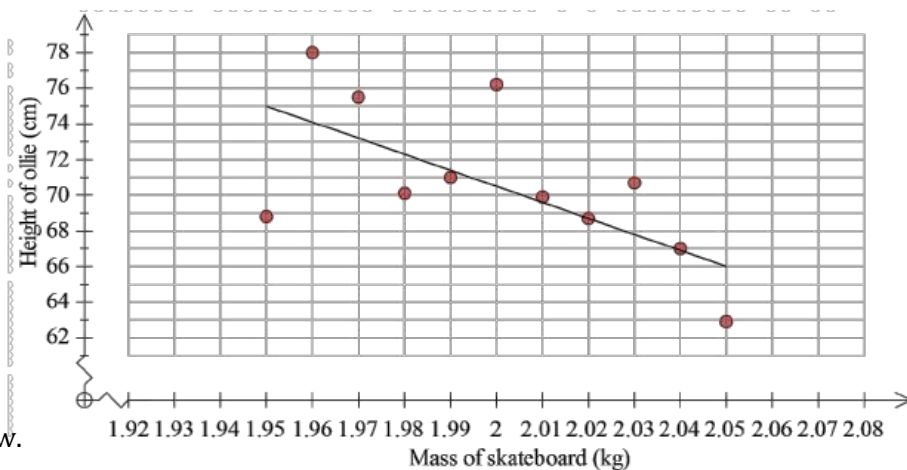
- 5 A survey was conducted between the number of hours studied by HSC students and how well they performed in the HSC. 2

The Pearson correlation coefficient r was found to be 0.9.

What conclusion can you make about the correlation between the number of hours studied by HSC students and how well they performed in the HSC?

$r = 0.9$ is a strong positive correlation. Based on the survey, students performed better in the HSC the more hours they studied.

- 6 The manufacturer of a brand of skateboards tested different skateboards to see if the mass of the skateboard affected how high a skateboarder could ollie (jump). The results are displayed in the scatterplot below.



- a. Identify the independent and dependent variables.

2

Independent variable = mass of skateboard

Dependent variable = height of ollie

- b. Describe the direction and strength of the relationship between the mass of a skateboard and the height of an ollie.

2

Direction = negative

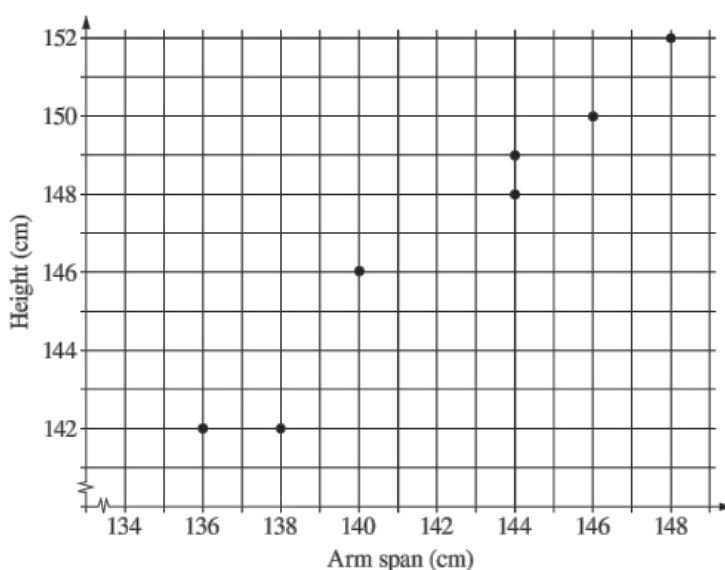
Strength = moderate

- c. Estimate the height of an ollie made with a skateboard that weighs 2.07 kg.

64 cm

1

- 7 A set of bivariate data is collected by measuring the height and arm span of seven children. The graph shows a scatterplot of these measurements.



- a. Calculate Pearson's correlation coefficient for the data, correct to two decimal places.

1

$$r \approx 0.98$$

- b. Identify the direction and the strength of the linear association between height and arm span.

1

Positive, strong

- c. The equation of the least-squares regression line is shown.

1

$$\text{Height} = 0.866 \times (\text{arm span}) + 23.7$$

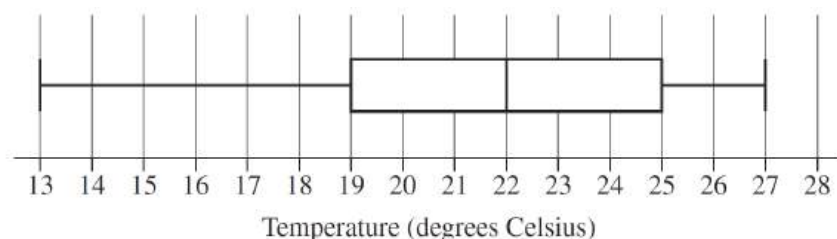
A child has an arm span of 143 cm. Calculate the predicted height for this child using the equation of the least-squares regression line.

$$\text{Height} = 0.866 \times 143 + 23.7 \approx 147.54 \approx 148 \text{ cm}$$

8 A cricket is an insect. The male cricket produces a chirping sound. A scientist wants to explore the relationship between the temperature in degrees Celsius and the number of cricket chirps heard in a 15-second time interval. Once a day for 20 days, the scientist collects data. Based on the 20 data points, the scientist provides the information below.

5

- A box-plot of the temperature data is shown.



- The mean temperature in the dataset is 0.525°C below the median temperature in the dataset.
- A total of 684 chirps was counted when collecting the 20 data points.

The scientist fits a least-squares regression line using the data (x, y) , where x is the temperature in degrees Celsius and y is the number of chirps heard in a 15-second time interval. The equation of this line is

$$y = -10.6063 + bx,$$

where b is the slope of the regression line. The least-squares regression line passes through the point $(\bar{x}, \bar{y})$ where $\bar{x}$ is the sample mean of the temperature data and $\bar{y}$ is the sample mean of the chirp data.

Calculate the number of chirps expected in a 15-second interval when the temperature is 19° Celsius. Give your answer correct to the nearest whole number.

$$\begin{aligned}\bar{x} &= 22 - 0.525 = 21.475, \quad \bar{y} = \frac{684}{20} = 34.2 \\ b &= \frac{y + 10.6063}{x} = \frac{\bar{y} + 10.6063}{\bar{x}} \quad \therefore \frac{y + 10.6063}{19} = \frac{34.2 + 10.6063}{21.475} \\ \therefore y &\approx 29\end{aligned}$$